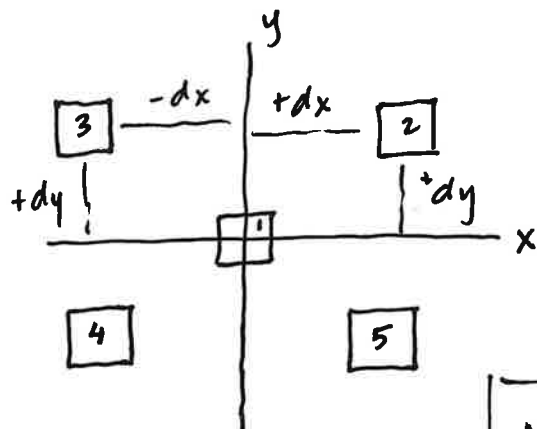


Interesting Product of Inertia (POI)

Example Problem 1

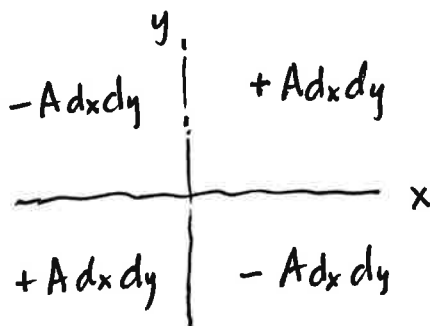


Five different Areas,
located differently in
x-y plane.

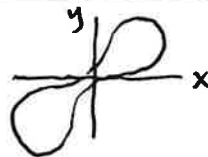
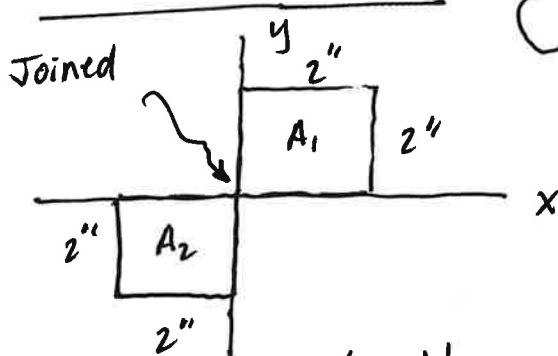
Determine I_{xy} for each area.

PAT! $I_{xy} = \bar{I}_{x'y'} + A \cdot dx \cdot dy$

No	A	$\bar{I}$	dx	dy	I_{xy}	= $A \cdot dx \cdot dy$
1	A	0	0	0	0	
2	A	0	+	+	+	
3	A	0	-	+	-	
4	A	0	-	-	+	
5	A	0	+	-	-	



Ex Problem 2



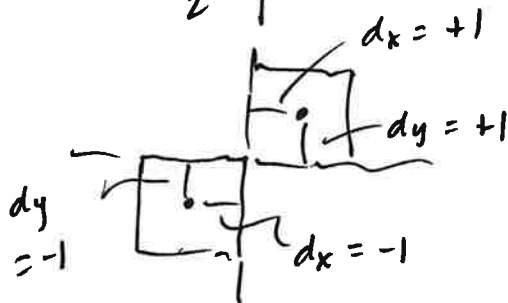
I_{xy} for this area:

$$I_{xy} = \sum (\bar{I} + A dx dy)$$

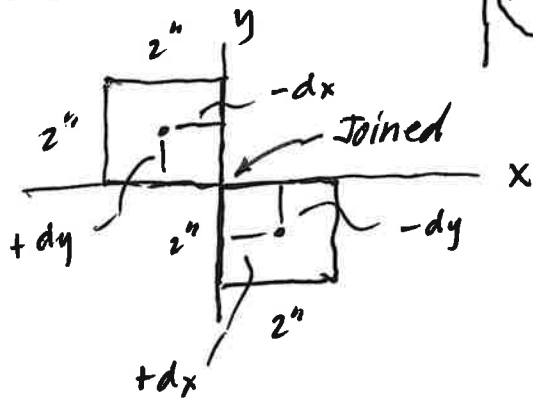
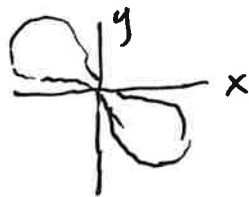
$$= [0 + (2)(2)(+1)(+1)]$$

$$+ [0 + (2)(2)(-1)(-1)]$$

$$I_{xy} = 4 + 4 = \boxed{+8 \text{ in}^4 = I_{xy}}$$



Ex Prob 3



$$I_{xy} = \sum (\bar{I} + A d_x d_y)$$

$$= [0 + (2)(2)(-1)(+1)]$$

$$+ [0 + (2)(2)(+1)(-1)]$$

$$I_{xy} = -4 - 4 = \boxed{-8 \text{ in}^4 = I_{xy}}$$